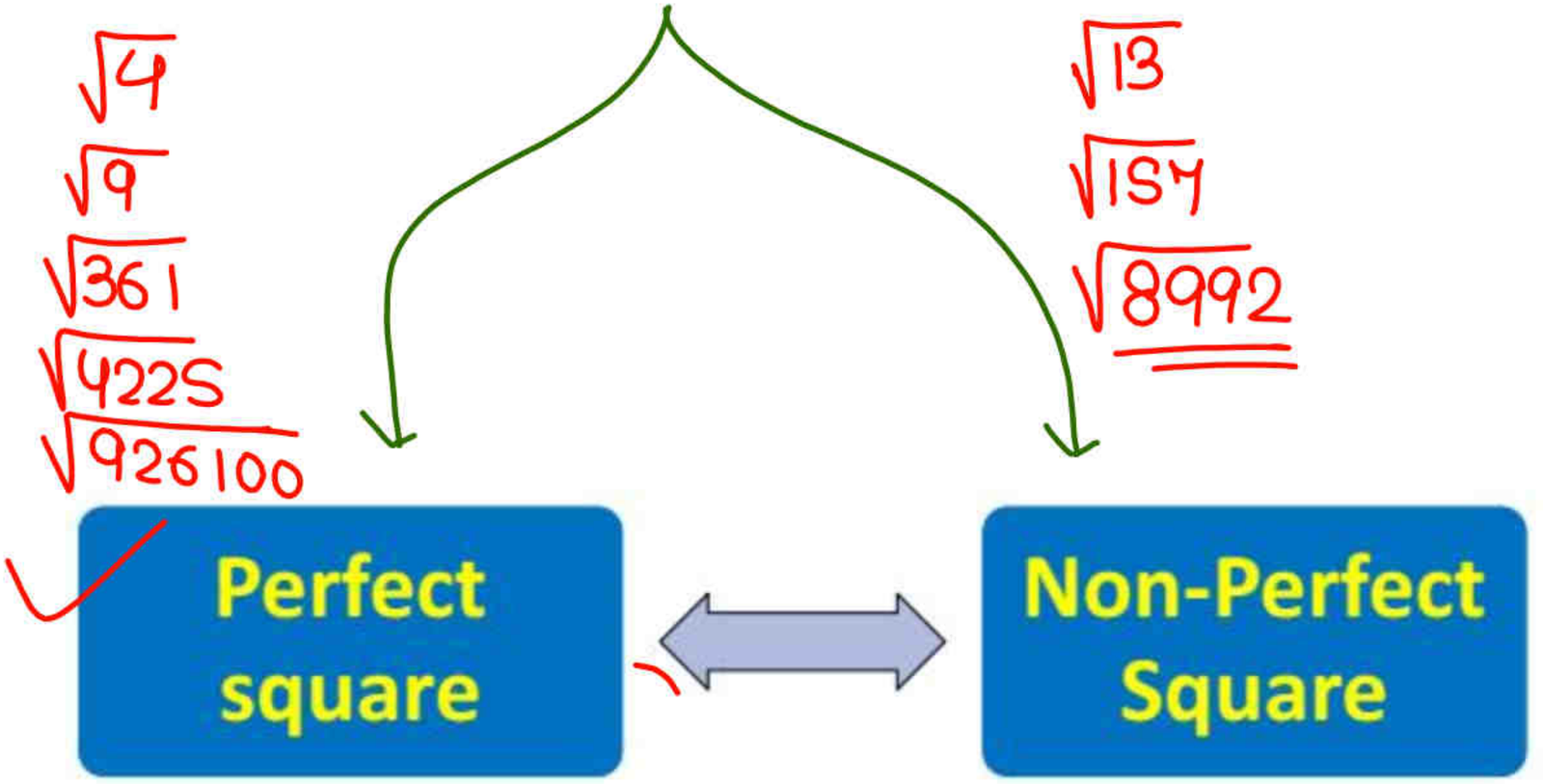
CALCULATION-03

SQUARE ROOT

(वर्ग मूल)

CLASS NOTES

Square Root / वर्गमूल



Square Root of perfect square
पूर्ण वर्ग का वर्गमूल

$1^2 = 1$

$2^2 = 4$

$3^2 = 9$

$4^2 = 16$

$5^2 = 25$

$6^2 = 36$

$7^2 = 49$

$8^2 = 64$

$9^2 = 81$

$10^2 = 100$

Example: Calculate the square root of 5184

Step 01: The unit digit of 5184 is 4, so its square root will be end on 2 or 8.

Step 02: Now leave two digits from end and check the remaining number. In this, case it is 51.

$$7^2 < 5 < 8^2$$

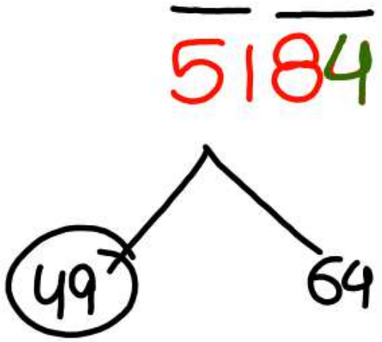
So, the tens digit of square root is 7.

Step 03: Now the square root of 5184 can be 72 or 78.

For this, we have to check whether 51 is closer to 7^2 or 8^2

As 51 is closer to 7^2.

Square root of 5184 is 72.



$\sqrt{2}$ ✓

or

$\sqrt{8}$

Example: Calculate the square root of 3136

Step 01: The unit digit of 3136 is 6, so its square root will be end on 4 or 6.

Step 02: Now leave two digits from end and check the remaining number. In this, case it is 31.

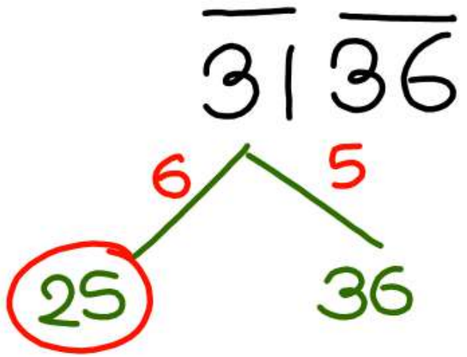
$$5^2 < 31 < 6^2$$

So, the tens digit of square root is 5.

Step 03: Now the square root of 3136 can be 54 or 56. For this, we have to check whether 31 is closer to 5^2 or 6^2

As 31 is closer to 6^2.

Square root of 3136 is 56.



5 4
or
5 6 ✓

Find the square root of

$(\overline{7744})$ = 82 or 88

64 | 13 | 4 | 81

$(\overline{1764})$ = 42 or 48

16 | 25

$(\overline{6561})$ = 81 or 89

64 | 81

$(\overline{7396})$ = 84 or 86

64 | 8 | 81

$\sqrt{2916} =$

$\sqrt{7569} =$

64 11 6 81

83 or 87

$\sqrt{4489} =$

$\sqrt{4225} =$

$\sqrt{9216} =$

11 8

81 100

94 or 96 ✓

$\sqrt{3249} =$

$\sqrt{1764} =$

$\sqrt{4624} =$

$\sqrt{9216} =$

Example: Calculate the square root of 15129

Step 01: The unit digit of 15129 is 9, so its square root will be end on 3 or 7.

Step 02: Now leave two digits from end and check the remaining number. In this, case it is 151.

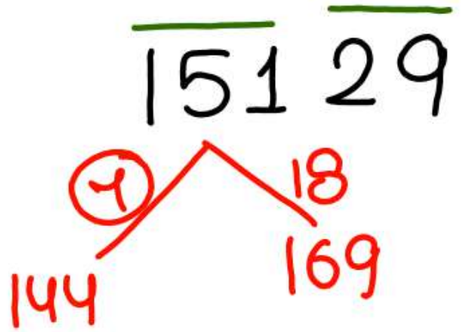
$$12^2 < 151 < 13^2$$

So, the first two digits of square root of 15129 is12.

Step 03: Now the square root of 15129 can be 123 or 127. For this, we have to check whether 151 is closer to 12^2 or 13^2

As 151 is closer to 12^2.

Square root of 15129 is 123.



12 3 or 12 4

$\sqrt{18769} =$ 133 or 137

169 9 196

$\sqrt{63001} =$ 251 or 259

625 676

$\sqrt{24964} =$

$\sqrt{70225} =$ 265

Ans

$\sqrt{54756}$ =

$\sqrt{33489}$ =

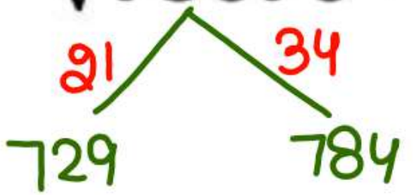
$\sqrt{69169}$ =

676 729

২৬ ৩ or ২৬৭

$\sqrt{75076}$ =

$\sqrt{75076} =$



274 or 276

Find the square root of $\overline{923521}$

$\overline{92}\,\overline{35}\,\overline{21}$ $= \hat{22} = 4$ ✓

-81

$\underline{9}\quad\underline{6}\quad\underline{1}$ → 7 → 49 ✓

$\underline{9}\quad\underline{6}\quad\underline{9}$ → 6 → 36 → 9 ✗

Find the square root of 173056

$\overline{1}\,\overline{73}\,\overline{05}\,\overline{6}$ = 4

4 | 1 | 4 → 9 → 9

or

4 | 1 | 6 → 2 → 4

Any in comment Box

Find the square root of 552049

Square Root of Non-perfect square
अपूर्ण वर्ग का वर्गमूल

$\sqrt{13}$ → 9 → y

↓

x

$$\frac{x+y}{2\sqrt{y}}$$

$$= \frac{13+9}{2\sqrt{9}}$$

$$= \frac{22}{2\times 3} = \frac{11}{3} = 3.66$$

Ex. $\sqrt{10}$ = → $9 \to y$

x

$$\frac{x+y}{2\sqrt{y}}$$

$$\frac{10+9}{2\times 3} = \frac{19}{6} = 3.1\ldots$$

Ex. $\sqrt{14}$ = $9 \to y$

$14 \to x$

$$\frac{x+y}{2\sqrt{y}}$$

$$\frac{14+9}{2\times 3} = \frac{23}{6} = 3.8\overline{3}$$

Ex. $\sqrt{18}$ = → 16 → y

x

$$\frac{x+y}{2\sqrt{y}}$$

$$= \frac{18+16}{2\times4} = \frac{34}{8} = 4.2\ldots$$

Ex. $\sqrt{29}$ = → 25

$$\frac{29+25}{2\times5} = \frac{5.4}{}$$

Ex. $\sqrt{57}$ = → 49

$$\frac{57+49}{2\times7} = \frac{106}{14} = 7.5$$

$\frac{x+y}{2\sqrt{y}}$

Ex. $\sqrt{265}$ = → 256 → y

x

$\frac{265+256}{2\times16} = \frac{521}{32} = 16.$

Ex. $\sqrt{374}$ = → 361

$\frac{374+361}{2\times19} = \frac{735}{38} = 19.3$

Ex. $\sqrt{445}$ =